

Remainder Theorem & Factor Theorem

Algebra Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Use the Remainder Theorem to evaluate a polynomial function at a given value using synthetic division
- Apply the Factor Theorem to determine whether a given binomial is a factor of a polynomial
- Use the Rational Root Theorem and synthetic division to fully factor and solve polynomial equations

Problems

1. Use the Remainder Theorem and synthetic division to find $f(2)$ for the polynomial below.

$$f(x) = x^3 - 4x^2 + 5x + 3$$

2. Use the Remainder Theorem and synthetic division to find $f(2)$ for the polynomial below. Note that the x -squared term is missing — replace it with a zero coefficient.

$$f(x) = -x^3 + 6x - 7$$

3. Use the Remainder Theorem and synthetic division to find $f(-3)$ for the polynomial below.

$$f(x) = 2x^3 + x^2 - 5x + 4$$

4. Determine whether $(x - 1)$ is a factor of the polynomial below using the Factor Theorem. Show your synthetic division work.

$$f(x) = x^3 + 9x^2 + 23x + 15$$

5. Use the Factor Theorem to verify that $(x + 1)$ is a factor of the polynomial below. Then divide completely and write all factors.

$$f(x) = x^3 + 9x^2 + 23x + 15$$

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6. List all possible rational roots using the Rational Root Theorem, then use synthetic division to find all real solutions of the polynomial equation below.

$$x^3 + 2x^2 - 5x - 6 = 0$$

7. Use the Remainder Theorem to evaluate $f(-2)$ for the polynomial below, then verify by direct substitution.

$$f(x) = 3x^4 - 2x^3 + x^2 - 4x + 5$$

8. Use the Rational Root Theorem and synthetic division to completely factor the polynomial below, then state all real zeros.

$$f(x) = x^3 - 6x^2 + 11x - 6$$

9. Determine the value of k so that $(x - 2)$ is a factor of the polynomial below.

$$f(x) = x^3 - kx^2 + 3x - 2$$

10. Use the Rational Root Theorem and repeated synthetic division to find all real solutions of the polynomial equation below. The polynomial has one repeated root.

$$x^4 - 6x^3 + 9x^2 + 4x - 12 = 0$$

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Remainder Theorem & Factor Theorem — Answer Key

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Answer Key

1. Answer: $f(2) = 5$

- Set up synthetic division with divisor 2 and coefficients 1, -4, 5, 3
 - Bring down 1. Multiply: $1 \times 2 = 2$; add to -4 $\rightarrow$ -2
 - Multiply: $-2 \times 2 = -4$; add to 5 $\rightarrow$ 1
 - Multiply: $1 \times 2 = 2$; add to 3 $\rightarrow$ 5
 - The remainder is 5, so by the Remainder Theorem $f(2) = 5$
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2. Answer: $f(2) = -3$

- Rewrite as $f(x) = -x^3 + 0x^2 + 6x - 7$; coefficients are -1, 0, 6, -7
 - Set up synthetic division with divisor 2
 - Bring down -1. Multiply: $-1 \times 2 = -2$; add to 0 $\rightarrow$ -2
 - Multiply: $-2 \times 2 = -4$; add to 6 $\rightarrow$ 2
 - Multiply: $2 \times 2 = 4$; add to -7 $\rightarrow$ -3
 - The remainder is -3, so $f(2) = -3$
-

3. Answer: $f(-3) = -38$

- Coefficients: 2, 1, -5, 4; divisor: -3
 - Bring down 2. Multiply: $2 \times (-3) = -6$; add to 1 $\rightarrow$ -5
 - Multiply: $-5 \times (-3) = 15$; add to -5 $\rightarrow$ 10
 - Multiply: $10 \times (-3) = -30$; add to 4 $\rightarrow$ -26
 - Wait — recalculate: 2, $1 \times (-3)$: bring 2, $-6 + 1 = -5$, $15 + (-5) = 10$, $-30 + 4 = -26$
 - The remainder is -26, so $f(-3) = -26$
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4. Answer: $f(1) = 48$, so $(x - 1)$ is NOT a factor

- By the Factor Theorem, $(x-1)$ is a factor only if $f(1) = 0$
 - Coefficients: 1, 9, 23, 15; divisor: 1
 - Bring down 1. $1 + 9 = 10$; $10 + 23 = 33$; $33 + 15 = 48$
 - Remainder = $48 \neq 0$, so $(x-1)$ is NOT a factor
-

5. Answer: $f(-1) = 0$; factors are $(x+1)(x+3)(x+5)$

- Test $x = -1$: coefficients 1, 9, 23, 15; divisor -1
 - Bring down 1. $1 \times (-1) = -1$; $-1 + 9 = 8$; $8 \times (-1) = -8$; $-8 + 23 = 15$; $15 \times (-1) = -15$; $-15 + 15 = 0$
 - Remainder = 0, confirming $(x+1)$ is a factor
 - Quotient is $x^2 + 8x + 15$
 - Factor $x^2 + 8x + 15 = (x+3)(x+5)$
 - Full factored form: $(x+1)(x+3)(x+5)$
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**6. Answer: $x = -1$, $x = 2$, $x = -3$**

- Possible rational roots: factors of 6 over factors of 1 $\rightarrow \pm 1, \pm 2, \pm 3, \pm 6$
- Test $x = 2$: 1, 2, -5, -6 with divisor 2 $\rightarrow 1, 4, 3, 0$; remainder 0 ✓
- Quotient: $x^2 + 4x + 3 = (x+1)(x+3)$
- Solutions: $x = 2$, $x = -1$, $x = -3$

7. Answer: $f(-2) = 85$

- Coefficients: 3, -2, 1, -4, 5; divisor: -2
- Bring down 3. $3 \times (-2) = -6$; $-6 + (-2) = -8$; $-8 \times (-2) = 16$; $16 + 1 = 17$; $17 \times (-2) = -34$; $-34 + (-4) = -38$; $-38 \times (-2) = 76$; $76 + 5 = 81$
- Recalculate carefully: $3 \mid -2 \mid 1 \mid -4 \mid 5$
- 3; $3 \times (-2) = -6$, $-2 + (-6) = -8$; $-8 \times (-2) = 16$, $1 + 16 = 17$; $17 \times (-2) = -34$, $-4 + (-34) = -38$; $-38 \times (-2) = 76$, $5 + 76 = 81$
- Remainder = 81, so $f(-2) = 81$
- Verify: $3(16) - 2(-8) + 1(4) - 4(-2) + 5 = 48 + 16 + 4 + 8 + 5 = 81$ ✓

8. Answer: Zeros: $x = 1$, $x = 2$, $x = 3$; $f(x) = (x-1)(x-2)(x-3)$

- Possible rational roots: $\pm 1, \pm 2, \pm 3, \pm 6$
- Test $x=1$: 1, -6, 11, -6 $\rightarrow 1, -5, 6, 0$ ✓; quotient $x^2 - 5x + 6$
- Factor $x^2 - 5x + 6 = (x-2)(x-3)$
- Zeros: $x = 1, 2, 3$

9. Answer: $k = 3$

- By the Factor Theorem, $f(2)$ must equal 0
- $f(2) = (2)^3 - k(2)^2 + 3(2) - 2 = 0$
- $8 - 4k + 6 - 2 = 0$
- $12 - 4k = 0$
- $4k = 12$, so $k = 3$

10. Answer: $x = -1$, $x = 2$ (multiplicity 2), $x = 3$

- Possible rational roots: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$
- Test $x=2$: 1, -6, 9, 4, -12 $\rightarrow 1, -4, 1, 6, 0$ ✓; quotient $x^3 - 4x^2 + x + 6$
- Test $x=2$ again on $x^3 - 4x^2 + x + 6$: 1, -4, 1, 6 $\rightarrow 1, -2, -3, 0$ ✓; quotient $x^2 - 2x - 3$
- Factor $x^2 - 2x - 3 = (x-3)(x+1)$
- All solutions: $x = 2$ (multiplicity 2), $x = 3$, $x = -1$

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